

YMCA Class Attendance & Scheduling System

Product Requirements Document (PRD) – Final Premium Edition

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1. Executive Summary

The YMCA Class Attendance & Scheduling System is a digital platform that replaces spreadsheet-based workflows with a unified mobile and web solution. The system includes an Instructor Mobile App, Manager Mobile App, and Manager Web Dashboard. It supports multi-branch operations, real-time attendance capture, and automated KPI reporting. The MVP delivery target is 3–4 weeks for the initial pilot branch, while the architecture is designed for future nationwide rollout.

2. Problem Statement

Branches currently rely on spreadsheets and manual processes to manage class schedules, instructor activity, and attendance counts. This approach leads to inconsistent data, missing or duplicated entries, limited historical analysis, and heavy administrative workload for instructors and managers. Leadership lacks timely and reliable insights into class performance, member engagement, and instructor utilization.

3. Project Objectives

The primary objectives of this project are to: (1) digitize attendance and scheduling; (2) standardize data collection across branches; (3) reduce manual administrative work; (4) improve reporting accuracy and timeliness; (5) provide managers and leadership with actionable KPIs; and (6) establish a scalable foundation for national deployment.

4. User Personas

Instructor – Teaches scheduled classes, needs quick mobile attendance entry, and the ability to edit submissions until manager approval.

Branch Manager – Oversees schedules, instructors, and attendance performance; requires web and mobile access to dashboards, approvals, KPIs, and exports.

5. MVP Scope

Included: multi-branch support; instructor mobile app; manager mobile app; manager web dashboard; scheduling module; headcount-based attendance; optional approval workflow; KPI dashboard based on the November 2025 model; PDF and Excel exports; Supabase backend; and YMCA-managed user accounts.

Excluded: member roster tracking, online registration, payments, digital waivers, wearable integration, and AI-driven predictive analytics (all candidates for future phases).

6. Functional Requirements

Instructor Mobile App

- Secure login using YMCA-provisioned credentials.
- Daily schedule view listing upcoming classes by time, location, and type.
- Headcount-only attendance entry with optional notes.
- Draft and Submit states, with ability to edit until approval.
- Prevention of duplicate submissions for the same scheduled class.

Manager Mobile App

- View daily schedule and attendance summaries.
- Approve or return instructor submissions.
- Quick KPI cards such as total attendance today and average per class.
- Access to instructor-level summaries.

Manager Web Dashboard

- Create and manage class schedules by date, time, branch, class type, and assigned instructor.
- Review attendance logs; filter by date, instructor, class type, or branch.
- Approve or request corrections when approval workflow is enabled.
- View KPI dashboards with monthly totals, averages, and instructor utilization.
- Export reports in PDF and Excel formats.

7. Scheduling

Branch managers define class schedules including branch, class type, class group (if used), instructor, start time, end time, and recurrence. The system must support one-off sessions and recurring patterns (e.g., every Tuesday/Thursday at 6 PM). Instructors may be assigned to multiple branches and classes.

8. Attendance

Attendance is captured as a single numeric headcount for each scheduled class instance. The instructor enters the total number of participants and may add an optional note. A single attendance record per class per date is enforced. Records progress through Draft → Submitted → Approved states when the approval workflow is enabled.

9. KPI Requirements

The system computes KPIs aligned to the November 2025 worksheet, including: Monthly Attendance Totals; Total Classes Held; Average Attendance per Class; Class Type Averages (e.g., Yoga, Cycling, Strength); Class Group Averages (e.g., Active Older Adult, Youth); Instructor Utilization (classes taught versus assigned); and attendance trends over days, weeks, and months.

10. Reporting & Exports

Managers can export monthly summary reports, class type and class group breakdowns, instructor activity reports, and detailed attendance lists. Exports are generated in Excel (.xlsx) and PDF. The monthly report package is designed to be presentation-ready for leadership, with summarized KPIs, charts, and narrative highlights.

11. Data Model Overview

The data model uses a normalized schema that includes: users, branches, instructors_branches (to support multi-branch assignments), class_types, class_groups, class_schedule, and attendance_logs. This structure supports secure role-based access, scalable reporting, and multi-branch operations.

12. User Stories

• As an instructor, I want to record attendance quickly so I can complete it during or immediately after class. • As an instructor, I want to edit my attendance entry until it has been approved. • As a branch manager, I want to build and maintain class schedules so that instructors always know what they are responsible for. • As a branch manager, I want to approve attendance submissions so that reported numbers are reliable. • As a branch manager, I want KPI dashboards and exports so that I can prepare monthly reports efficiently.

13. Non-Functional Requirements

- Performance: Attendance submission should complete in under two seconds under normal load.
- Reliability: Target uptime of at least 99% for core services.
- Security: Supabase row-level security and encryption for sensitive data.
- Scalability: Architecture must support growth to multiple branches and eventually a national footprint.
- Data Retention: Attendance and schedule data retained for at least seven years.

14. Release Roadmap

Phase 1 – MVP (3–4 Weeks): Implement instructor mobile app, manager mobile app, scheduling, attendance, optional approval workflow, and basic reporting. Phase 2: Deliver enhanced KPI dashboards, trend analytics, and expanded reporting capabilities. Phase 3: Introduce national-level dashboards, cross-branch comparisons, and potential AI-based forecasting and optimization.

Appendix A – KPI Definitions

This appendix formally defines each KPI, including:

- Monthly Attendance Total = Sum of all approved attendance headcounts in the selected month.
- Average Attendance per Class = $\text{Monthly Attendance Total} \div \text{Number of classes held}$.
- Class Type Average = $\text{Total attendance for a given class type} \div \text{Number of sessions of that type}$.
- Class Group Average = $\text{Total attendance for a class group} \div \text{Number of sessions in that group}$.
- Instructor Utilization = $(\text{Number of classes taught by instructor} \div \text{Number of classes assigned}) \times 100$.
- Trend Metrics = Attendance values aggregated by day, week, or month, compared over time.

Appendix B – Wireframe Placeholders

Wireframe placeholders describe the intended layout of key screens: • Instructor Daily Schedule – List of classes by time with a call-to-action to enter attendance. • Instructor Attendance Entry – Simple form with headcount field, optional notes, and Submit/Draft controls. • Manager Mobile Dashboard – KPI cards, pending approvals list, and today's schedule. • Manager Web Dashboard – Tabbed layout for KPIs, schedule, attendance logs, and exports. These will be translated into high-fidelity UI designs by the UX team.

Appendix C – Data Dictionary (High Level)

Key entities: • users: id, name, email, role (instructor, manager). • branches: id, name, region. • instructors_branches: instructor_id, branch_id. • class_types: id, name. • class_groups: id, name. • class_schedule: id, branch_id, class_type_id, group_id, instructor_id, start_time, end_time, recurrence. • attendance_logs: id, schedule_id, instructor_id, headcount, notes, status, created_on. A full technical data dictionary can be produced as a follow-on artifact.

Appendix D – Release Roadmap Detail

Phase 1 (3–4 Weeks): Focus on core flows—scheduling, attendance entry, approvals, and export of basic reports. Phase 2 (6–8 Weeks): KPI dashboards, time-series analytics, and richer exports. Phase 3 (8–12 Weeks): National dashboards, benchmarking across branches, and advanced optimization features.

Appendix E – User Flows

Instructor Flow: Login → View today's classes → Select class → Enter headcount → Add notes (optional) → Save as Draft or Submit → (If enabled) Manager Approval → Record marked as Approved. Manager Flow: Login → View dashboard KPIs → Open approvals queue → Review attendance details → Approve or Return for edit → Export monthly reports as needed.

Appendix F – Future Enhancements

Potential future enhancements include: integration with member rosters; member-level attendance tracking; online registration and waitlisting; capacity and space planning; predictive analytics for class demand; and automated notifications and reminders for instructors and members.